Soham Amol Dube

Email: ssohamdube@gmail.com — Phone: +91-7775939391 — LinkedIn — GitHub

Profile Summary

AI-focused Computer Science undergraduate (Expected 2027) with experience in financial time-series modeling, NLP, and intelligent automation. Built distributed data pipelines using Apache Spark and developed LSTM-based predictive systems for stock movement forecasting. Demonstrated measurable ML performance improvements and ranked Top 10% nationally in Shell Hackathon. Interested in deploying responsible AI solutions to enhance operational efficiency in large-scale banking environments.

Technical Skills

Programming: Python, SQL, JavaScript, C++
AI/ML: LSTM, NLP, Transformers, Scikit-learn, XGBoost, LightGBM
Big Data: Apache Spark
Cloud: AWS (EC2, S3, CloudWatch)
Frameworks: MERN Stack (MongoDB, Express, React, Node.js)
Tools: Git, Power BI, Tableau, REST APIs

Experience

Tech Team Member – Health-O-Tech Jan 2025 – Present

- Contributing to AI-enabled healthcare workflow solutions to improve operational efficiency.
- Supporting feature testing, debugging, and validation to ensure reliable system performance.
- Collaborating with cross-functional teams to integrate scalable automation enhancements.

MERN Stack Developer Intern (IIT Ropar – Prof. Sudarshan Iyengar) Dec 22, 2025 – Jan 18, 2026

- Translated case study requirements into full-stack MERN solutions for workflow digitization.
- Designed RESTful APIs and optimized database queries, improving response time by 25%.
- Implemented secure authentication and modular architecture for scalable deployment.

IBM Generative AI Intern May 2025 – Jun 2025

- Developed NLP classification pipelines using pseudo-labeling and data augmentation.
- Improved F1-score from 84% to 88% through model refinement and validation testing.
- Evaluated AI performance reliability and analyzed adoption challenges in operational workflows.

Projects

AI-Based Stock Movement Prediction using Spark + LSTM

- Processed historical stock data using Apache Spark for distributed feature engineering.
- Engineered financial indicators including RSI, MACD, Bollinger Bands, Moving Averages, CCI, and Stochastic Oscillator.
- Built LSTM model (PyTorch) using 10-time-step sequences to predict price movement direction.
- Evaluated model performance across train/test splits to assess predictive stability and accuracy.

Fuel Blend Prediction – Shell Hackathon (Top 10% Nationally)

- Processed 100K+ sensor records and developed ensemble ML models for predictive optimization.
- Reduced inference runtime to under 5 minutes, enabling near real-time decision support.

Education

B.Tech in Computer Science Engineering(Artificial Intelligence and Machine Learning)
VIT Bhopal University Aug 2023 – May 2027
CGPA: 8.43

Achievements

- Top 10% National Rank – Shell Hackathon
- Top 5% – NPTEL - Cloud Computing Certification
- IIT Ropar - Internship Certificate